

Replication Project

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Applied Stats II

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Who Shot the Bullets? Exposure to Violence and Attitudes Toward Peace: Evidence from the 2016 Colombian Referendum

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Historical Context

- ▶ Colombian Civil War
- ▶ FARC, paramilitaries, BACRIM
- ▶ Election of Juan Manuel Santos and Peace Talks
- ▶ 2016 Peace Referendum

Research Question and Hypotheses

Question: Does variation in experiences of violence also translate into different attitudes toward peace? Specifically, investigates whether exposure to violence by the FARC, as opposed to violence by the paramilitaries and BACRIM, had an observable impact on vote choice.

Hypotheses:

- ▶ In areas where Group A has committed most of the violence, residents are more likely to support a peace agreement with Group A.
- ▶ In areas where B has committed most of the violence, residents are less likely to support a peace agreement with Group A.

Independent and Dependent Variables

Dependent Variable:

- ▶ % of votes per municipality supporting the 2016 peace agreement referendum

Independent Variables:

Set of variables measuring the average number of violent attacks carried out by FARC and paramilitaries/BACRIM from the years 1988-2010.

- ▶ (ln) FARC attacks (ln_atfarc_cede)
- ▶ (ln) Paramilitary/BACRIM attacks (ln_atparabacrim_cede)

Control Variables included in Regression

Set of variables included in their models that could have an impact on perceived violence and voting in referendum

- ▶ % Votes for Santos in 2014 Election (ref_partic)
- ▶ % Participation in Referendum (share_santos_2_100)
- ▶ Poverty Measure (incidence_multidim_pov_2005_100)
- ▶ Population Measure (lpop)
- ▶ Rurality Index (indrural2005)
- ▶ Coca Cultivation (cultivated100)
- ▶ Oil Present (oil)
- ▶ Elevation (ln_altitud)
- ▶ Education (coberturabrutaeduc)

Departmental Effects

Fixed Effects Model

- ▶ Control for unobserved characteristics that do not change over time and are specific to each department
- ▶ 32 departments and 1 capital district

OLS Model

```
# find which reference category matches stata's for model 7
for (ref in all_depts) {
  dept_vars_try <- paste(setdiff(all_depts, ref), collapse = " + ")
  f_try <- as.formula(paste("ref_si ~ ln_atfarc_cede + ln_atparabacrim_cede +
    ref_partic + share_santos_2_100 +
    incidence_multidim_pov_2005_100 + lpop + indrural2005 +
    cultivated100 + oil + ln_altitud +
    cobeturabrutaeduc +", dept_vars_try))

  model_try <- lm(f_try, data = df)
  intercept <- coef(model_try)["(Intercept)"]
  if (abs(intercept - 18.78) < 0.01) {
    cat("Model 7 reference category:", ref, "\n")
  }
}
```

```
# remove reference department
dept_vars_7 <- paste(setdiff(all_depts, "departamento_24"), collapse = " + ")

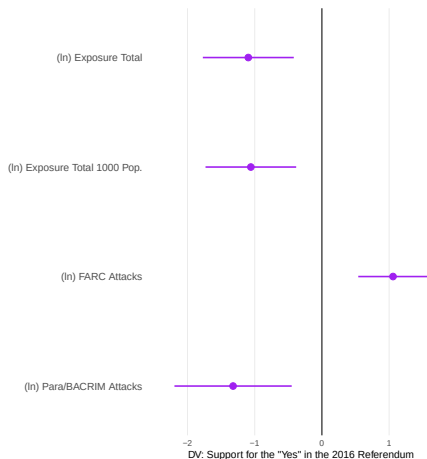
# run model 7
f7 <- as.formula(paste("ref_si ~ ln_atfarc_cede + ln_atparabacrim_cede +
  ref_partic + share_santos_2_100 +
  incidence_multidim_pov_2005_100 + lpop + indrural2005 +
  cultivated100 + oil + ln_altitud +
  cobeturabrutaeduc +", dept_vars_7))

model_7 <- lm(f7, data = df)
summary(model_7)
```

Table 1. OLS Identity Perpetrator (Models 5-7)

	(5)	(6)	(7)
(ln) FARC attacks	1.100*** (0.249)	1.050*** (0.254)	1.058*** (0.264)
(ln) Paramilitary/BACRIM attacks	-1.608*** (0.418)	-1.470*** (0.431)	-1.322*** (0.445)
% Support Santos 2014	0.596*** (0.022)	0.589*** (0.023)	0.585*** (0.025)
Rural Index	0.092*** (0.021)	0.083*** (0.022)	0.085*** (0.023)
Coca		2.022** (1.028)	2.128** (1.039)
Oil		-1.627 (1.036)	-1.781* (1.047)
Department Fixed Effects	✓	✓	✓
Constant	30.221** (12.045)	20.008*** (7.414)	18.778* (9.602)
Num.Obs.	927	887	857
R2	0.720	0.710	0.699
R2 Adj.	0.708	0.698	0.685

Figure 1. Exposure to Violence and Vote Choices in the Referendum



Interpreting Results and Findings

Table 1:

- ▶ Holding all other variables constant, on average, for every one percent increase in the mean number of FARC attacks in a municipality, the percentage of votes per municipality supporting the 2016 peace agreement referendum increases by .01058%.
- ▶ Holding all other variables constant, on average, for every one percent increase in the mean number of Para/BACRIM attacks in a municipality, the percentage of votes per municipality supporting the 2016 peace agreement referendum decreases by .01322%.

Figure 1:

- ▶ "Overall levels of violent attacks by all armed groups in each Colombian municipality do not have an impact on citizens' support for the peace agreement, clear (and statistically significant) effects emerge when we disaggregate by the identity of the perpetrator."

Problems with Replication

- ▶ Authors did not use a uniform reference category
- ▶ Some departments had a very small number of municipalities within them
- ▶ Authors had an incorrect interpretation of their regression coefficients

My Contribution: Multilevel Model

- ▶ Extension of linear regression, but allows for more accurate inferences for grouped data by allowing the intercept, or both the intercept and slopes, to vary
- ▶ Group and individual level variation is considered within the estimation of the group-level coefficients
- ▶ In between no pooling and full pooling ¹

¹Gelman & Hill (2018)

Multilevel Model Setup

```
# need to create a new column with the departments together and factor it
df$departamento <- factor(
  names(df[, all_depts])[max.col(df[, all_depts])])
```

```
# model 7: farc attacks, para/BACRIM attacks, % participation in ref,
# % support santos, poverty, population, rural index, coca, oil, elevation,
# education coverage
multi_model_7 <- lmer(ref_si ~ ln_atfarc_cede + ln_atparabacrim_cede +
  ref_partic + share_santos_2_100 +
  incidence_multidim_pov_2005_100 +
  lpop + indrural2005 + cultivated100 + oil + ln_altitud
  + coberturaabrutaeduc + (1 | departamento),
  data = df)
summary(multi_model_7)
```

Table 2. Multilevel Model Results

	Model 5	Model 6	Model 7
(Intercept)	15.63*	18.26**	20.16*
	(6.66)	(6.88)	(8.46)
ln_atfarc_cede	1.05***	0.98***	1.03***
	(0.24)	(0.25)	(0.26)
ln_atparabacrim_cede	-1.67***	-1.45***	-1.34**
	(0.41)	(0.42)	(0.44)
share_santos_2_100	0.60***	0.59***	0.58***
	(0.02)	(0.02)	(0.02)
indrural2005	0.09***	0.08***	0.08***
	(0.02)	(0.02)	(0.02)
Num. obs.	927	887	857
Num. groups: departamento	33	27	26
Var: departamento (Intercept)	19.71	19.02	19.47
Var: Residual	78.73	79.81	79.97

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

Multilevel Model Intercepts

```
# determine and show shifted intercepts  
data.frame(coef(multi_model_7)$departamento)[1]
```

Output:

Intercept.	
departamento_10	24.27479
departamento_11	21.28139
departamento_12	17.28407

departamento_10: The average expected value of the percentage of votes per municipality within department 10 supporting the 2016 peace agreement referendum, when all covariates are 0 is 24.27%.

Understanding the Incorrect Interpretation

(ln) FARC attacks	1.100***	1.050***	1.058***
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Author's Interpretation: "If the mean of FARC attacks in a given municipality increases by 1%, the yes vote in the referendum increases by approximately 1.06 points."

- ▶ They cannot make this claim about their outcome variable
- ▶ Why?

Predicted Probabilities

In order to understand if this claim was correct and how/why they could be making it, I chose to determine predicted probabilities of % yes votes in referendum for varying numbers of FARC attacks and para/BACRIM attacks, and subsequently plot these values against the exponentiated ($\ln$) FARC Attacks and ($\ln$) Para/BACRIM Attacks variable values to get the true raw counts.

Predicted Probabilities

```
# repeat the first 100 rows of the original dataframe
prediction_data_para <- df[rep(1, 100), ]

# fill all department columns with 0
dept_cols <- grep("^departamento_", names(prediction_data_para), value = TRUE)
prediction_data_para[, dept_cols] <- 0

# fill every other variable, make all values equal to the mean
prediction_data_para$ln_atfarc_cede <- mean(df$ln_atfarc_cede, na.rm = TRUE)
prediction_data_para$ref_partic <- mean(df$ref_partic, na.rm = TRUE)
prediction_data_para$share_santos_2_100 <- mean(df$share_santos_2_100, na.rm = TRUE)

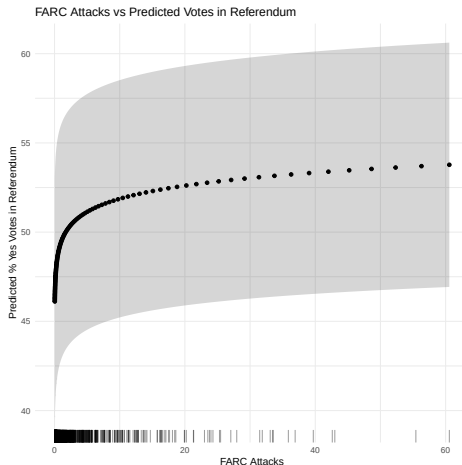
# have FARC attacks variable vary amongst ranges between the min and max values
prediction_data_para$ln_atparabacrim_cede <- as.numeric(seq(min(df$ln_atparabacrim_cede, na.rm = TRUE),
max(df$ln_atparabacrim_cede, na.rm = TRUE),
length.out = 100))

# predict new data using model_7 on the prediction data created
preds_para <- predict(model_7, newdata = prediction_data_para, type = "response",
interval = "confidence")

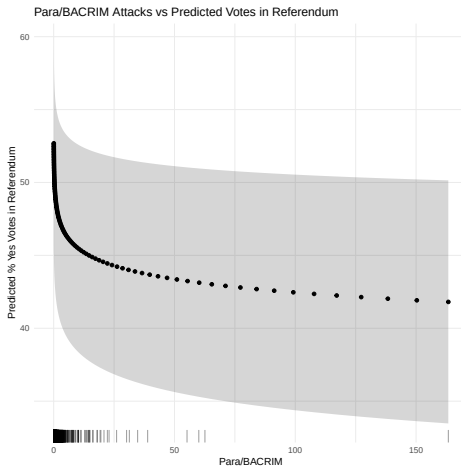
# unlist predicted column
prediction_data_para$predicted <- preds_para[, "fit"]

# confidence intervals around predicted probabilities
prediction_data_para$lower_95 <- preds_para[, "lwr"]
prediction_data_para$upper_95 <- preds_para[, "upr"]
```

FARC Attack Counts vs Predicted Probabilities of % Yes Votes in Referendum



Para/BACRIM Attack Counts vs Predicted Probabilities of % Yes Votes in Referendum



What this Means

The author's interpretation is wrong, the correct interpretation is:

FARC Attacks: Holding all other variables constant, for every 1% increase in the mean log of FARC attacks in a given municipality, the yes vote in the referendum increases by approximately 0.0106 points.

Para/BACRIM Attacks: Holding all other variables constant, for every 1% increase in the mean log of para/BACRIM attacks in a given municipality, the yes vote in the referendum decreases by approximately 0.0132 points.

References

Gelman, A., & Hill, J. (2018). *Data analysis using regression and multilevel/hierarchical models*. Cambridge University Press.

Clark, M. (n.d.). *Models demystified – Michael Clark*.

Github.io. Retrieved March 25, 2026, from
<https://m-clark.github.io/category=mixed%20models>